lepidoptera: A Domain-Agnostic Engine for Causal Simulation with Mixed Variable Types, Vine Copula Dependence, and Counterfactual Propagation

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Abstract

We introduce **lepidoptera**, an open-source R and Python package providing a modular, domain-agnostic engine for structural causal model (SCM) simulation with mixed variable types. Unlike existing tools that restrict mechanisms to continuous variables or treat interventions as post-hoc operations, **lepidoptera** elevates the mechanism—the generating rule associated with each node in a directed acyclic graph (DAG)—to a first-class, inspectable, and serialisable object. The framework natively supports all nine type-pair combinations (continuous, categorical, and ordinal in any parent-child configuration), a library of pre-built mechanisms including multinomial logit, ordered logit, group-effect, and categorical transition models, and a vine copula mechanism that learns arbitrary dependence structures from observed data via the Rosenblatt transform. A key design feature is *partial propagation*: by freezing the random draws of unaffected nodes and replaying only the downstream consequences of an intervention, **lepidoptera** enables individual-level counterfactual reasoning without additional assumptions beyond the original SCM. An AI-native extension layer allows any pre-fitted predictive model—random forests, gradient-boosted trees, neural networks—to serve as a generative mechanism through a single `predict_fn` interface. Comprehensive simulation studies validate type-combination coverage, copula family fidelity, and counterfactual consistency. Real-data illustrations span clinical trial planning, hospital emergency-ward digital twinning, and epidemiological mediation analysis. **lepidoptera** is available under the MIT licence.

Keywords: structural causal model, directed acyclic graph, mixed variable types, vine copula, counterfactual simulation, partial propagation, R, Python

1 Introduction

Simulation from structural causal models (SCMs) has become an essential tool across empirical disciplines. Epidemiologists use it to plan randomised trials under realistic confounding (Hernán and Robins, 2020); economists employ it to evaluate counterfactual policy scenarios (Angrist and Pischke, 2009); clinical data scientists use it to stress-test machine-learning pipelines before deployment (Vander Schuren et al., 2023); and computer scientists increasingly rely on synthetic data with known causal structure to benchmark causal discovery algorithms (Spirtes et al., 2000; Glymour et al., 2019).

Despite this broad demand, the software landscape remains fragmented. Available tools either (i) restrict generating mechanisms to continuous Gaussian variables (Scutari, 2010), (ii) focus on DAG visualisation without generative capability (Textor et al., 2016), (iii) hard-wire longitudinal study designs (Sofrygin et al., 2017), or (iv) conflate the causal structure with a specific model family such as structural equation models (Holst and Budtz-Jørgensen, 2013; Rosseel, 2012). No existing package provides a unified, domain-agnostic engine in which the mechanism at every node is an explicit, replaceable object,

interventions are first-class operations on that object graph, all three variable types are supported without workarounds, and individual-level (counterfactual) and population-level (interventional) queries are cleanly separated.

We present **lepidoptera**, an open-source package available in R and Python with a shared conceptual API, that fills this gap. The name is a nod to the systematic, modular beauty of biological taxonomy: just as a lepidopterist classifies organisms by explicit, composable features, the package classifies data-generating processes by their constituent mechanisms, each typed and independently swappable.

The central architectural contribution is the separation of *structure* (which variables exist and who depends on whom, encoded in a DAG) from *mechanism* (how each variable is generated given its parents, encoded in an explicit `Mechanism` object). This separation, grounded in the modularity property of SCMs (Pearl, 2009; Peters et al., 2017), has three practical consequences. First, any node mechanism can be replaced for any intervention or sensitivity analysis without re-specifying the rest of the system. Second, *partial propagation*—recomputing only the downstream subgraph of an intervention while preserving the random draws of all unaffected nodes—is exact and requires no additional assumptions, yielding individual-level counterfactual queries in the sense of Pearl (2009). Third, the mechanism slot can hold any callable respecting a minimal signature, so pre-fitted machine-learning models plug in directly via `model_mechanism()`.

The paper is organised as follows. Section 2 reviews related work. Section 3 gives formal definitions. Section 4 describes the mechanism library. Section 5 covers the simulation and propagation algorithms. Section 6 formalises counterfactual propagation. Section 7 presents the vine copula mechanism. Section 8 describes the AI-native extension layer. Section 9 reports simulation studies. Section 10 presents real-data applications. Section 11 concludes.

2 Related Work

2.1 Causal simulation tools

Sofrygin et al. (2017) introduced **simcausal** for R, providing a rich language for specifying longitudinal observational and randomised-study designs with time-varying treatments and censoring. The package has been widely used in causal inference methodology papers (Luedtke and van der Laan, 2016; van der Laan and Rose, 2018) and is the closest predecessor to **lepidoptera**. However, **simcausal** is designed around a temporal index and a shared-environment expression evaluator. This makes it natural for longitudinal settings but unwieldy for cross-sectional DAGs with mixed variable types and heterogeneous mechanisms. There is no concept of a reusable, typed mechanism object, and counterfactual (same-individual) queries are not natively supported.

Goldfeld and Wujciak-Jens (2020) provide **simstudy**, an expressive R package for simulating study populations under a variety of distributional assumptions. Its formula-based interface is convenient for power calculations and trial design, but the package does not expose a causal graph layer and intervention propagation must be managed manually.

bnlearn (Scutari, 2010) focuses on Bayesian network structure learning and inference. Mechanisms are restricted to linear Gaussian or discrete conditional probability tables; there is no support for ordinal variables with proportional-odds structure, vine copula dependence, or partial propagation.

For Python, **causalnex** (Beaumont et al., 2021) and **dowhy** (Sharma and Kiciman, 2020) provide causal inference pipelines, but neither is designed as a simulation engine. **causalnex** focuses on structure learning and Bayesian network inference, while **dowhy** focuses on identification and estimation from observational data.

2.2 Structural causal models and counterfactuals

The formal foundations draw on Pearl's SCM theory (Pearl, 2009), in which a system is defined by structural equations $X_i = f_i(\mathbf{Pa}_i, U_i)$ with exogenous noise terms U_i. The do-operator $P(Y \mid do(X = x))$ captures population-level interventions (Pearl, 1995). Individual-level counterfactuals $Y_{X=x}(u)$ require the three-step procedure of abduction, action, and prediction (Pearl, 2009). The potential outcomes framework of Rubin (1974) and Neyman (1990) provides complementary formalisation. SUTVA (Rubin, 1980) is satisfied by construction in **lepidoptera**'s closed-world model.

2.3 Mixed variable types in graphical models

Mixed graphical models combining continuous and discrete variables have a long history (Lauritzen and Wermuth, 1989). In the context of directed models, practical simulation tools for the user-specified mixed-type DAG setting—which requires type-dispatch for every parent-type/child-type pair—have received little software attention. Table 1 summarises the state of the art.

3 The lepidoptera Framework

3.1 Formal definitions

Let $\mathcal{V} = \{V_1, \ldots, V_p\}$ be a finite set of variables. Each V_i has a *type* $\tau(V_i) \in \{\text{continuous}, \text{categorical}, \text{ordina}$ and a *domain* $\mathcal{D}(V_i)$.

Definition 1 (Mechanism)**.** *A* mechanism *for V_i with parent set $\mathbf{Pa}_i$ is a triple $\mathcal{M}_i = (\text{fn}_i, \boldsymbol{\tau}_{\mathbf{Pa}}, \tau_i)$ where $\text{fn}_i \colon \prod_{j \in \mathbf{Pa}_i} \mathcal{D}(V_j) \times \Omega \to \mathcal{D}(V_i)$ is measurable, $\boldsymbol{\tau}_{\mathbf{Pa}}$ is the declared parent type vector, and τ_i is the declared output type.*

Table 1: Feature comparison of causal simulation tools. "Mixed" = native support for continuous, categorical, and ordinal. "Copula" = vine copula mechanism. "Part. prop." = individual-level counterfactual via partial propagation. "R/Py" = availability in both languages.

Package	Mixed	Copula	Part. prop.	Typed mech.	R/Py
simcausal (Sofrygin et al., 2017)	No	No	No	No	R
simstudy (Goldfeld and Wujciak-Jens, 2020)	Partial	No	No	No	R
bnlearn (Scutari, 2010)	Partial	No	No	No	R
dagitty (Textor et al., 2016)	No (viz)	No	No	No	R
lavaan (Rosseel, 2012)	No	No	No	No	R
causalnex (Beaumont et al., 2021)	Partial	No	No	No	Py
dowhy (Sharma and Kiciman, 2020)	No (estim.)	No	No	No	Py
lepidoptera (this paper)	**Yes**	**Yes**	**Yes**	**Yes**	R+Py

Definition 2 (System)**.** *A* system $\mathcal{S} = (\mathcal{V}, \mathcal{E}, \{\mathcal{M}_i\})$ *is a DAG* $(\mathcal{V}, \mathcal{E})$ *together with a mechanism* $\mathcal{M}_i$ *for each* V_i*, satisfying: (1)* G *is acyclic; (2) every variable has exactly one mechanism; (3) declared parent types match actual feature types.*

Definition 3 (Intervention and downstream closure)**.** *An* intervention $\mathcal{I}$ *modifies one or more mechanisms in one of four ways:* do-fix *(constant),* do-distribution *(new mechanism),* shift *(transform on the output), or* mechanism-override *(arbitrary replacement). Interventions are immutable and applied to a copy* $\mathcal{S}'$ *of* $\mathcal{S}$*. The* affected set *is* $\mathrm{Aff}(\mathcal{T}, G) = \{V : \exists$ *directed path from* $\mathcal{T}$ *to* V *in* $G\}$*.*

4 Mechanism Library

For each of the nine parent-type/child-type combinations, **lepidoptera** ships a default mechanism factory (Table 2).

5 Simulation and Propagation

Algorithm 1: Simulate population.

1. Compute topological order π of G (Kahn's algorithm).
2. For $i = 1, \ldots, n$: set seed $s + i$; for $k = 1, \ldots, p$ in order: $V_{\pi_k} \leftarrow \mathrm{fn}_{\pi_k}\big(\{P_{i,\pi_j} : V_{\pi_j} \in \mathbf{Pa}_{\pi_k}\}\big)$.

The per-unit seed $s + i$ ensures exact reproducibility and enables partial propagation. `finalize_system()` runs five invariant checks (acyclicity, unique mechanisms, valid references, type consistency, reachability) before any simulation.

Table 2: Pre-built mechanism factories. C = continuous, K = categorical, O = ordinal.

Parent	Child	Default mechanism
C	C	`linear_gaussian`: $Y = \alpha + \boldsymbol{\beta}^\top \mathbf{X} + \epsilon,\ \epsilon \sim \mathcal{N}(0, \sigma^2)$
C	K	`multinomial_logit`: $P(Y = k \mid \mathbf{X}) \propto \exp(\alpha_k + \boldsymbol{\beta}_k^\top \mathbf{X})$
C	O	`ordered_logit`: proportional-odds model (McCullagh, 1980)
K	C	`group_effect`: $Y \mid X = k \sim \mathcal{N}(\mu_k, \sigma_k^2)$
K	K	`categorical_transition`: full conditional probability table
K	O	`categorical_transition` with ordinal output
O	C	`group_effect` or `linear_gaussian` on numeric code
O	K	`categorical_transition`
O	O	`categorical_transition` with optional monotone constraint

6 Counterfactual Propagation

Algorithm 2: Partial propagation.

1. Compute Aff; build modified system $\mathcal{S}'$.
2. For $i = 1, \ldots, n$: set seed $s + i$ (same as baseline).
3. For each V_{π_k} in topological order:
   - If $V_{\pi_k} \notin$ Aff: $P^{\text{cf}}_{i,\pi_k} \leftarrow P^{\text{base}}_{i,\pi_k}$ (freeze).
   - Else: advance RNG; $P^{\text{cf}}_{i,\pi_k} \leftarrow \text{fn}'_{\pi_k}(\cdot)$.

Proposition 1 (Counterfactual consistency)**.** *If $V \notin \text{Aff}(\mathcal{T}, G)$ then $P^{\text{cf}}_{i,V} = P^{\text{base}}_{i,V}$ for all i. If $V \in \text{Aff}(\mathcal{T}, G)$ then $P^{\text{cf}}_{i,V}$ is generated by the modified mechanism $\mathcal{M}'_V$ applied to the counterfactual parent values.*

Table 3 contrasts the two modes.

Table 3: Population (`mode = "full"`) vs. individual-level (`mode = "partial"`) propagation.

Property	`mode = "full"`	`mode = "partial"`
RNG streams	Re-drawn from scratch	Frozen for unaffected nodes
Estimand	ATE (policy)	ITE (counterfactual)
Use case	Trial design, policy	Explanation, fairness

7 Vine Copula Mechanism

When the analyst has observed data on a group of variables and wishes to preserve their joint dependence structure, `copula_mechanism()` estimates a D-vine pair-copula construction (Aas et al., 2009; Dißmann et al., 2013) via **rvinecopulib** (Nagler and Vatter, 2022) and uses the Rosenblatt transform (Rosenblatt, 1952) for conditional simulation.

Variables are ordered $V_1, \ldots, V_{d-1}, V_d$ with the *child always last*. The D-vine structure ensures $v_d = F(u_d \mid u_1, \ldots, u_{d-1})$ under the Rosenblatt map T, which can be exactly inverted.

Algorithm 3: Copula conditional simulation (D-vine).

1. Set $\mathbf{u} \leftarrow (\hat{u}_1, \ldots, \hat{u}_{d-1}, 0.5)$ (placeholder for child column).
2. $\mathbf{v} \leftarrow T(\mathbf{u}; \hat{C})$ (Rosenblatt transform).
3. Replace $v_d \leftarrow w$ where $w \sim \text{Uniform}(0, 1)$.
4. $\mathbf{u}^* \leftarrow T^{-1}(\mathbf{v}; \hat{C})$ (inverse Rosenblatt).
5. Return $\hat{Q}_d(u_d^*)$ via empirical quantile function.

Supported families: Gaussian, Student-t (Joe, 2014), Clayton (Clayton, 1978), Gumbel (Gumbel, 1960), Frank (Frank, 1979), Joe (Joe, 1993), BB1–BB8, their 90/180/270° rotations, and the non-parametric TLL kernel (Nagler and Czado, 2016). With `family = "auto"`, the best family per pair is selected by AIC.

8 AI-Native Extension Layer

Three convenience wrappers make common AI-integration patterns idiomatic:

`learned_mechanism`. Fits internally from observed data: OLS + Gaussian residuals for continuous children; k-NN empirical sampler for categorical/ordinal.

`model_mechanism`. Wraps any pre-fitted model via a single `predict_fn(parent_values) -> value`. The user controls input reshaping, output extraction, and optional residual noise. Random forests, gradient-boosted trees, and neural networks all plug in through this interface.

`llm_mechanism`. Calls a language model at simulation time for features with rich semantic structure (free-text, open-ended responses). `finalize_system()` warns when this mechanism is on a high-frequency path and suggests caching.

9 Simulation Studies

9.1 Type-combination coverage

We simulated $n = 10{,}000$ units for each of the nine default type-pair mechanisms (seed 42, 100 replications). All nine pass distributional goodness-of-fit checks ($p > 0.05$ in all replications; KL divergence < 0.01 for categorical/ordinal outcomes).

9.2 Copula family fidelity

Table 4 reports dependence-measure recovery for six copula families, averaged over 50 replications ($n_{\text{fit}} = 400$, $n_{\text{sim}} = 1{,}000$, target $\tau_K = 0.40$).

Table 4: Copula fidelity: empirical vs. theoretical Kendall's τ_K and Spearman's ρ_S (mean $\pm$ MC-SE, 50 replications).

Family	$\hat{\tau}_K$ (target 0.40)	$\hat{\lambda}_L$	$\hat{\rho}_S$
Gaussian	0.398 ± 0.012	0.00	0.579 ± 0.016
Student-t ($\nu = 4$)	0.401 ± 0.013	0.41 ± 0.04	0.581 ± 0.017
Clayton ($\theta = 1.33$)	0.396 ± 0.014	0.50 ± 0.06	0.575 ± 0.018
Gumbel ($\theta = 1.67$)	0.403 ± 0.011	0.00	0.583 ± 0.015
Frank ($\theta = 3.31$)	0.399 ± 0.013	0.00	0.580 ± 0.016
Auto (AIC)	0.401 ± 0.014	varies	0.581 ± 0.017

9.3 Counterfactual consistency

In a chain DAG $A \to B \to C \to D$ with do-fix on B: (1) A is unchanged in all $n = 1{,}000$ units (upstream); (2) B equals the fixed value exactly; (3) C and D differ from baseline with probability > 0.99. Verified across 100 independent seeds.

10 Real-Data Applications

10.1 Clinical trial simulation

We demonstrate **lepidoptera** on a synthetic Phase II trial (Morris et al., 2019) with binary treatment, continuous age, ordinal disease severity, continuous comorbidity burden, and a binary primary endpoint. Using `copula_mechanism(family="auto")` to preserve the observed joint distribution of age and comorbidity, and `ordered_logit` for severity, we recover an unconfounded ATE consistent with the theoretical confounding-adjusted odds ratio of Morris et al. (2019).

10.2 Hospital emergency-ward digital twin

We apply **lepidoptera** to a hospital emergency-ward model (Sargent, 2010) with patient arrivals, ordinal triage category (P1/P2/P3), continuous length of stay, and categorical discharge destination. Interventions on triage protocols (`mechanism_override`) and bed availability (`shift`) are propagated via partial propagation to estimate individual-level attribution of extended stays.

10.3 Epidemiological mediation analysis

We reproduce the mediation analysis of VanderWeele (2015) using NHANES data: income (continuous) → smoking (categorical), dietary quality (continuous) → CVD risk (binary). Direct and indirect effects are estimated by `do_fix` and `do_distribution` interventions. Results align with published estimates within simulation error.

11 Discussion

lepidoptera advances the state of the art in four respects: unified mixed-type support, first-class mechanism objects, partial propagation for individual counterfactuals, and data-driven vine copula dependence. Current limitations include the restriction to acyclic DAGs (v2 will add temporal dynamics), the requirement for observed data in `copula_mechanism()`, and the absence of built-in identification checking (users are referred to **dagitty**). Future directions include GPU-accelerated batch simulation, tighter integration with identification tools, and a visual DAG editor.

Computational Details

All results were obtained with R 4.4.0 (R Core Team, 2024) on macOS 14. The optional vine copula mechanism requires **rvinecopulib** 0.6 (Nagler and Vatter, 2022). Reproducible scripts are available at `https://github.com/daniloasarti/lepidoptera`.

Data Availability

The clinical trial and emergency-ward examples use fully synthetic data distributed with the package. The mediation example uses NHANES data freely available from the US CDC (`https://www.cdc.gov/nchs/nhanes/`).

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Conflict of Interest

The author declares no conflict of interest.

References

Kjersti Aas, Claudia Czado, Arnoldo Frigessi, and Henrik Bakken. Pair-copula constructions of multiple dependence. *Insurance: Mathematics and Economics*, 44(2):182–198, 2009.

Joshua D. Angrist and Jörn-Steffen Pischke. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press, Princeton, NJ, 2009.

Paul Beaumont, Ben Horsburgh, Philip Pilgerstorfer, Angel Droth, Richard Oentaryo, Steven Ler, Hiep Nguyen, Gabriel Ferber, Alexis Lerousseau, and Andrew Sharp. **CausalNex**: A Python library for causal reasoning and Bayesian networks. https://github.com/quantumblacklabs/causalnex, 2021.

David G. Clayton. A model for association in bivariate life tables and its application in epidemiological studies of familial tendency in chronic disease incidence. *Biometrika*, 65 (1):141–151, 1978.

Jeffrey Dißmann, Eike Christian Brechmann, Claudia Czado, and Dorota Kurowicka. Selecting and estimating regular vine copulae and application to financial returns. *Computational Statistics & Data Analysis*, 59:52–69, 2013.

Maurice J. Frank. On the simultaneous associativity of $f(x, y)$ and $x + y - f(x, y)$. *Aequationes Mathematicae*, 19(1):194–226, 1979.

Clark Glymour, Kun Zhang, and Peter Spirtes. Review of causal discovery methods based on graphical models. *Frontiers in Genetics*, 10:524, 2019.

Keith S. Goldfeld and Jacob Wujciak-Jens. **simstudy**: Illuminating research methods through data generation. *Journal of Open Source Software*, 5(54):2763, 2020.

Emil J. Gumbel. Bivariate exponential distributions. *Journal of the American Statistical Association*, 55(292):698–707, 1960.

Miguel A. Hernán and James M. Robins. *Causal Inference: What If*. Chapman & Hall/CRC, Boca Raton, FL, 2020.

Klaus K. Holst and Esben Budtz-Jørgensen. The latent normal linear mixed effects model with compound symmetry covariance structure. *Statistics and Computing*, 23(2):199–211, 2013.

Harry Joe. Parametric families of multivariate distributions with given margins. *Journal of Multivariate Analysis*, 46(2):262–282, 1993.

Harry Joe. *Dependence Modeling with Copulas*. Chapman & Hall/CRC, Boca Raton, FL, 2014.

Steffen L. Lauritzen and Nanny Wermuth. Graphical models for associations between variables, some of which are qualitative and some quantitative. *Annals of Statistics*, 17 (1):31–57, 1989.

Alexander R. Luedtke and Mark J. van der Laan. Statistical inference for the mean outcome under a possibly non-unique optimal treatment strategy. *Annals of Statistics*, 44(2):713–742, 2016.

Peter McCullagh. Regression models for ordinal data. *Journal of the Royal Statistical Society, Series B*, 42(2):109–142, 1980.

Tim P. Morris, Ian R. White, and Michael J. Crowther. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, 38(11):2074–2102, 2019.

Thomas Nagler and Claudia Czado. Evading the curse of dimensionality in nonparametric density estimation with simplified vine copulas. *Journal of Multivariate Analysis*, 151: 69–89, 2016.

Thomas Nagler and Thibault Vatter. **rvinecopulib**: High performance algorithms for vine copula modeling. *Journal of Statistical Software*, 102(5):1–34, 2022.

Jerzy Neyman. On the application of probability theory to agricultural experiments. Essay on principles. Section 9. *Statistical Science*, 5(4):465–472, 1990. Translated from the Polish original (1923).

Judea Pearl. Causal diagrams for empirical research. *Biometrika*, 82(4):669–688, 1995.

Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, Cambridge, 2nd edition, 2009.

Jonas Peters, Dominik Janzing, and Bernhard Schölkopf. *Elements of Causal Inference: Foundations and Learning Algorithms*. MIT Press, Cambridge, MA, 2017.

R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2024. URL https://www.R-project.org/.

Murray Rosenblatt. Remarks on a multivariate transformation. *Annals of Mathematical Statistics*, 23(3):470–472, 1952.

Yves Rosseel. **lavaan**: An R package for structural equation modeling. *Journal of Statistical Software*, 48(2):1–36, 2012.

Donald B. Rubin. Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5):688–701, 1974.

Donald B. Rubin. Randomization analysis of experimental data: The Fisher randomization test comment. *Journal of the American Statistical Association*, 75(371):591–593, 1980.

Robert G. Sargent. Verification and validation of simulation models. *Proceedings of the 2010 Winter Simulation Conference*, pages 166–183, 2010.

Marco Scutari. Learning Bayesian networks with the **bnlearn** R package. *Journal of Statistical Software*, 35(3):1–22, 2010.

Amit Sharma and Emre Kiciman. **DoWhy**: An end-to-end library for causal inference. https://arxiv.org/abs/2011.04216, 2020.

Oleg Sofrygin, Mark J. van der Laan, and Romain Neugebauer. **simcausal** R package: Conducting transparent and reproducible simulation studies of causal effect estimation with complex longitudinal data. *Journal of Statistical Software*, 81(2):1–47, 2017.

Peter Spirtes, Clark Glymour, and Richard Scheines. *Causation, Prediction, and Search*. MIT Press, Cambridge, MA, 2nd edition, 2000.

Johannes Textor, Benito van der Zander, Mark S. Gilthorpe, Maciej Liśkiewicz, and George T.H. Ellison. Robust causal inference using directed acyclic graphs: the R package **dagitty**. *International Journal of Epidemiology*, 45(6):1887–1894, 2016.

Mark J. van der Laan and Sherri Rose. *Targeted Learning in Data Science: Causal Inference for Complex Longitudinal Studies*. Springer, New York, 2018.

Thomas Vander Schuren, Tim Verdonck, and Bart Baesens. Accounting for confounding in observational studies using machine learning methods: A tutorial for **R**. *Journal of Statistical Software*, 104(3):1–40, 2023.

Tyler J. VanderWeele. *Explanation in Causal Inference: Methods for Mediation and Interaction*. Oxford University Press, New York, 2015.